AMAZON NOVA HACKATHON 2026 · SUBMISSION DESCRIPTION

AutoApply

AI-Powered Job Application Agent

Amazon Nova Act | Nova 2 Lite · Bedrock | Nova 2 Sonic · Bedrock | Python 3.11 · FastAPI | Next.js 16 | UI Automation | Agentic AI | Voice AI

Project Summary

AutoApply is an AI agent fleet that automates the full job application pipeline — from search to submission to interview preparation — built on all three Amazon Nova services. A candidate uploads their resume once, sets a job title and location, and the system takes over.

The project was purpose-built for the pan-African tech market, where a single job application takes 25–45 minutes to complete manually and the average candidate submits 60–80 applications before receiving an offer. AutoApply reduces that 45-minute process to a 45-second approval.

3	3×	45s	0
Nova services integrated	parallel browser agents	per application (vs 45 min)	credentials needed to demo

1 How AutoApply Leverages Amazon Nova

Nova Act
nova.amazon.com

Powers two distinct agent types: **job search agents** that open Indeed, LinkedIn, and Glassdoor in parallel browser sessions and extract structured listings using Pydantic schema output; and **application agents** that navigate to a job URL, fill every form field with the candidate's profile, upload the resume via Playwright interop, answer screening questions, and submit — all while streaming a live action feed over WebSocket. Nova Act's natural language instruction model handles diverse, non-standardised form layouts across global and African job boards without any site-specific scripting. Human escalation is implemented via `devtools_frontend_url` for CAPTCHA and high-stakes review gates.

Nova 2 Lite
Amazon Bedrock

Used for all language intelligence tasks via the Bedrock `converse` API: **resume parsing** (PDF → structured skills/experience/contact JSON); **job match scoring** (0–100 score, keyword overlap, ATS pass rate, experience level); **skill gap coaching** (explains how to reframe existing experience to address missing requirements, with specific resume bullet rewrites); **tailored cover letter generation** (company-specific, cliché-free, under 350 words); and **screening question answering** using only verified information from the candidate's profile.

Nova 2 Sonic

Amazon Bedrock

Drives a real-time voice interview coaching session accessible directly from any job card. The frontend captures microphone audio as 16kHz 16-bit mono PCM, streams it over WebSocket to the FastAPI backend, which relays it to Nova 2 Sonic via Bedrock's bidirectional streaming API (`invoke_model_with_bidirectional_stream`). Spoken AI responses are streamed back and played through the browser's AudioContext. Nova 2 Sonic is given a system prompt constructed from the specific job description and the candidate's resume summary, producing role-specific interview questions and spoken feedback after each answer.

All three Nova services are used for ***distinct, non-overlapping purposes*** *— Nova Act for UI interaction, Nova 2 Lite for language reasoning, and Nova 2 Sonic for real-time voice. None could substitute for the others. The full application runs end-to-end with graceful mock fallbacks when API credentials are not configured.*

2 Key Features

Parallel Job Search
3 Nova Act agents searching Indeed, LinkedIn & Glassdoor simultaneously via Python threading

Intelligent Matching
Nova 2 Lite scores every listing 0–100 with ATS analysis, skill overlap, and experience level

Skill Gap Coaching
Specific bullet rewrites that reframe existing experience — not generic advice

Tailored Cover Letters
Company-specific, cliché-free letters generated per application by Nova 2 Lite

Automated Form Submission
Nova Act handles diverse layouts, file uploads, and screening Q&A with human review gate

Voice Interview Coach
Nova 2 Sonic conducts role-specific mock interviews with live spoken feedback

Real-Time Agent Feed
WebSocket stream shows every agent action live — transparent, auditable automation

Human-in-the-Loop
CAPTCHA escalation + review-before-submit toggle; candidate stays in control

3 Community Impact

AutoApply was built explicitly for the pan-African tech community. Developers in Lagos, Nairobi, and Accra have comparable skills to peers in San Francisco but spend a disproportionate share of their time on application mechanics — because the tools that exist were priced for Western markets ($30–50/month USD) and built for Western job board layouts.

The code is open source (MIT). Any developer can deploy it with their own AWS credentials. Community organisations such as She Code Africa (10,000+ members) and the Andela Learning Community can run shared instances. Post-hackathon, African-specific boards — Jobberman, BrighterMonday, Fuzu, Careers24 — will be added as additional Nova Act agent threads. A commercial tier would be priced in local currency at purchasing-power-adjusted rates.

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4 Technical Stack

Backend: Python 3.11 · FastAPI · SQLAlchemy + PostgreSQL · Amazon S3 (resume storage) · Boto3 (Bedrock) · nova-act SDK · Alembic migrations
Frontend: Next.js 16 · TypeScript · Tailwind CSS · NextAuth.js (Google / Amazon / Apple / Facebook OAuth) · Axios · AudioContext (PCM capture)
Infrastructure: Docker Compose · Amazon Bedrock · Amazon S3